



BRENT OZAR
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Running SQL Server in AWS & Azure

Deck 1 p1

Abstract

You're used to managing your own SQL Servers in VMs, but now they're asking you to work in the cloud.

You're already comfortable installing, backing up, and tuning "regular" SQL Servers, and you just need to catch up with what's different about the cloud.

You don't want hype - you just want honest opinions from someone who doesn't work for a cloud vendor.



Deck 1 p2

What you'll learn today

How to assess your current workloads

How to pick the right VM type for your workload

How to size storage for performance, not size

How to assess wait stats in cloud VMs

How to back up directly to cloud-native file systems

Where other cloud database services make sense

Deck 1 p3



This material expires.

This stuff is true as of fall 2022.

Things change continuously.

I do update the material twice a year,
but if something changed yesterday...

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Some of my attendees



Deck 1 p5



Alex the Architect Antelope

May be data architect or app architect

Has a team of 5+ developers

Focused on 1 application, 1 database

Possibly doing Software-as-a-Service,
but each client is in their own database

Is probably here today

Deck 1 p6





Alex the Architect asks...

Can I get by without hiring a full time DBA?

Does the cloud DB work well with .NET?

Can we easily add performance capacity?

Do I need to worry about running out of space?

Deck 1 p7



Billie the Small Business Bird

Manages systems for a small business

3-4 SQL Servers, a few dozen databases

Most of the databases are third party

Has 1-2 sysadmins, but no full time DBA

Probably isn't here – but their consultant is

Deck 1 p8



Billie the Small Biz Bird asks



Can we get a cloud database that all of my third party vendors will accept?

Can we set it up and forget it?

Will someone else manage backups for me?

Deck 1 p9



Elliot the Enterprise Elephant

Dozens of SQL Servers

Thousands of databases

Multiple in-house dev teams

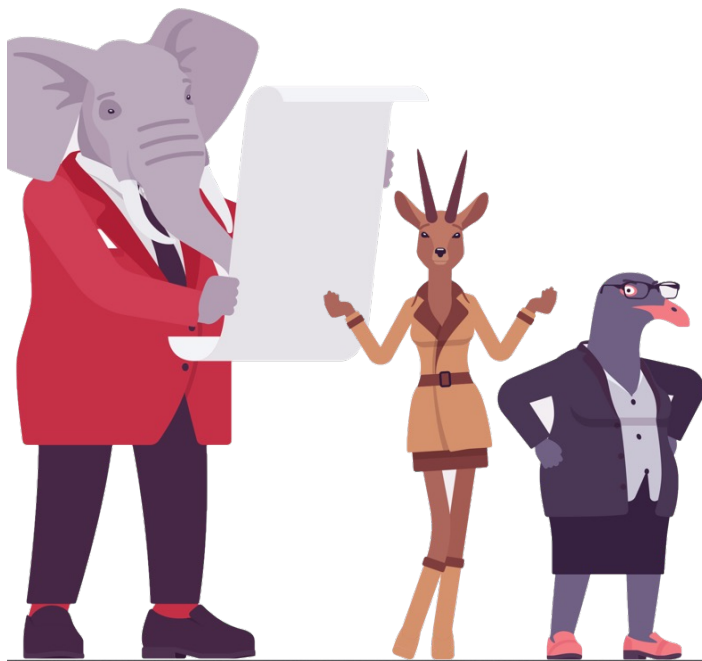
A dedicated full time DBA team

Corporation with compliance rules

Is here today making a strategy

Deck 1 p10





I'll use 'em
throughout
the session.

Deck 1 p11



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How to Assess Your Workloads

Deck 1 p12

Agenda

The 3 metrics to gather

Small/medium/large ranges for each metric

The 3 questions to ask stakeholders

Designing your own performance test

Deck 1 p13



3 migration metrics

3 migration metrics

1. How much data are we hosting?
**Total data size in GB/TB,
and number of databases.**
2. How fast are we going?
Batch Requests per second.
3. How hard are we working to achieve that speed?
**Wait time ratio over time
(like hours per wait in a 1-hour time span)**

Deck 1 p15



3. Wait Time Ratio

If the ratio is...	That means...
Zero	The server isn't waiting on anything.
0.25	In 1 second, queries wait 0.25 seconds on stuff. In 1 hour, queries wait 15 minutes on stuff.
0.5	In 1 hour, queries wait 30 minutes on stuff.
1.0	In 1 hour, queries wait 1 hour on stuff.
2.0	In 1 hour, queries wait 2 hours on stuff. (We have multiple queries waiting all the time.)

Deck 1 p16



How each user gathers this data



Deck 1 p17

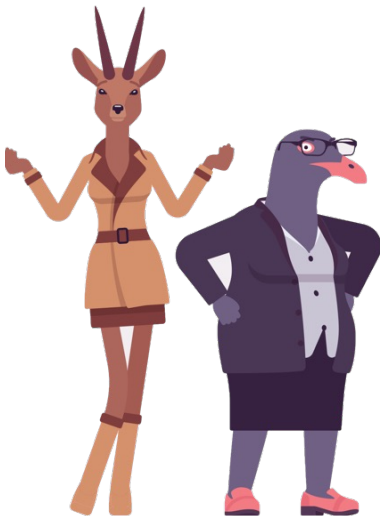


How they gather it

Alex the Antelope Architect and Billie the Small Business Bird rarely have SQL-specific monitoring.

They download the First Responder Kit:

<http://firstresponderkit.org>



Deck 1 p18



Gathering the metrics

Install in any database, and then:

```
1 sp_BlitzFirst @Seconds = 60
```

200 %					
Results Messages					
	Priority	FindingsGroup	Finding	URL	Details
1	0	sp_BlitzFirst 2022-0...	From Your Community Volunteers	http://FirstRe...	<?ClickToSeeDetails -- We hope you found this ...
2	50	Query Problems	Compilations/Sec High	https://www....	<?ClickToSeeDetails -- Number of batch request...
3	50	Query Problems	High Percentage Of Runnable Queries	https://enikda...	<?ClickToSeeDetails -- On the 1st pass, 45.00% ...
4	50	Query Problems	Statistics Updated Recently	https://www....	<?ClickToSeeDetails -- In the last 15 minutes, st...
5	100	Query Performance	Queries with 10000x cardinality misestimations	https://www....	<?ClickToSeeDetails -- The query on SPID 69 h...
6	200	Wait Stats	SOS_SCHEDULER_YIELD	https://www....	<?ClickToSeeDetails -- For 896 seconds over th...
7	200	Wait Stats	LCK_M_S	https://www....	<?ClickToSeeDetails -- For 878 seconds over th...
8	200	Wait Stats	PREEMPTIVE_OS_QUERYREGISTRY	https://www....	<?ClickToSeeDetails -- For 114 seconds over th...
9	200	Wait Stats	ASYNC_NETWORK_IO	https://www....	<?ClickToSeeDetails -- For 79 seconds over the ...
10	200	Wait Stats	WAIT_ON_SYNC_STATISTICS_REFRESH	https://www....	<?ClickToSeeDetails -- For 61 seconds over the ...

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10	200	Wait Stats	WAIT_ON_SYNC_STATISTICS_REFRESH	https://www....	<?ClickToSeeDetails -- For 61 seconds over the ...
11	250	Server Info	Batch Requests per Sec	https://www....	<?ClickToSeeDetails -- 41.97 -- ?>
12	250	Server Info	CPU Utilization	https://www....	<?ClickToSeeDetails -- 43%. Ring buffer details:...
13	250	Server Info	Wait Time per Core per Sec	https://www....	<?ClickToSeeDetails -- 8.74 -- ?>
14	251	Server Info	Database Count	https://www....	<?ClickToSeeDetails -- 2 -- ?>
15	251	Server Info	Database Size, Total GB	https://www....	<?ClickToSeeDetails -- 299.89111328124 -- ?>
16	251	Server Info	Memory Grant/Workspace info	https://www....	<?ClickToSeeDetails -- Grants Outstanding: 11 ...
17	255	Thanks!	From Your Community Volunteers	http://FirstRe...	<?ClickToSeeDetails -- To get help or add your ...

Sampling it over time

Run this in an SQL Agent job every 15 minutes, and it'll automatically keep 7 days of history:

```
EXEC sp_BlitzFirst  
    @OutputDatabaseName = 'dbatools',  
    @OutputSchemaName = 'dbo',  
    @OutputTableName = 'BlitzFirst';
```

Get median, max data for your peak workload times.

Deck 1 p21



Elliot has enterprise monitoring software.

Monitors all servers, not just SQL

Gathers OS-level metrics, Perfmon, sometimes some SQL Server DMVs

Has nice GUI to access different servers, troubleshoot them live

Deck 1 p22





This rarely works.

Rarely tracks total data size per server, Batch Req/sec, wait time.

Even when the info is there, it's hard to get out a single inventory for all servers.

Elliot often has to resort to the First Responder Kit too.

Deck 1 p23



Size ranges for each metric

What's normal for SQL Server?

Microsoft doesn't publish metrics.

We do from SQL ConstantCare®:

<https://www.brentozar.com/sql-constantcare/>

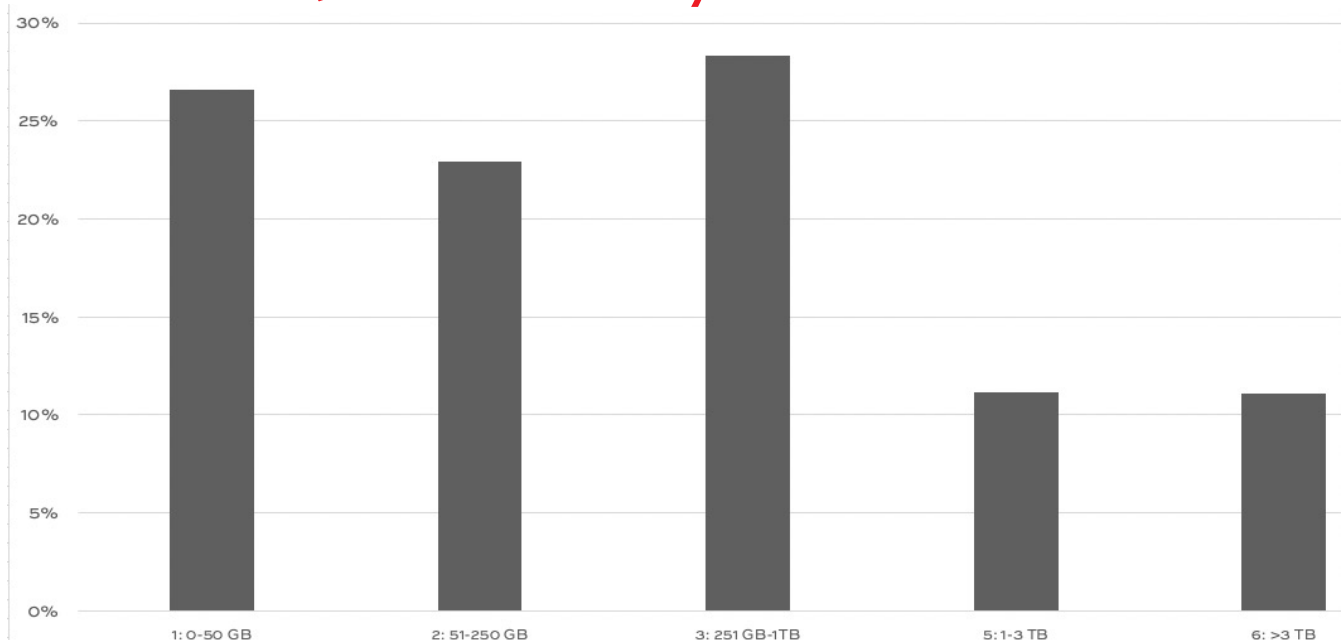
These metrics are biased:

- ~400 users, about 3,500 SQL Servers
- Only servers people care enough to monitor
- Only people who know about BrentOzar.com
- Tends to be smaller businesses, not enterprises

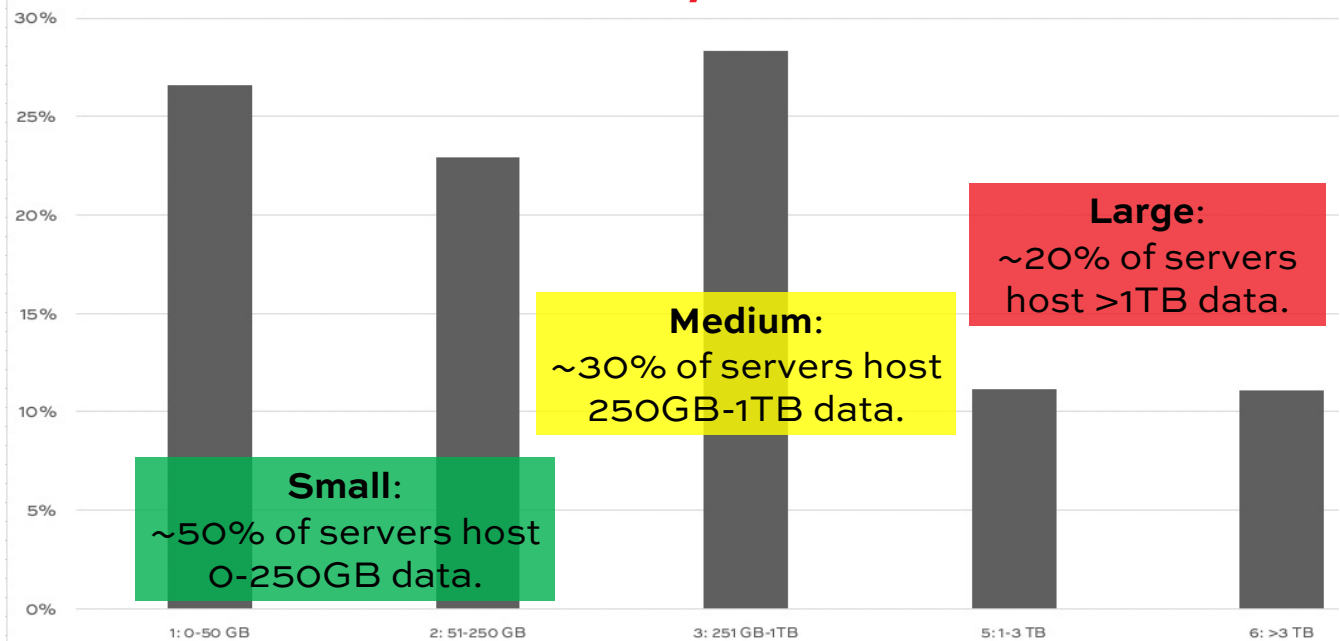
Deck 1 p25



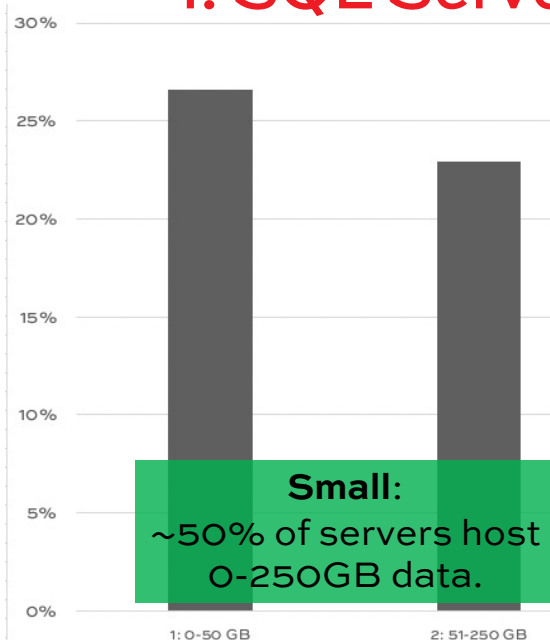
1. SQL Servers by total data size



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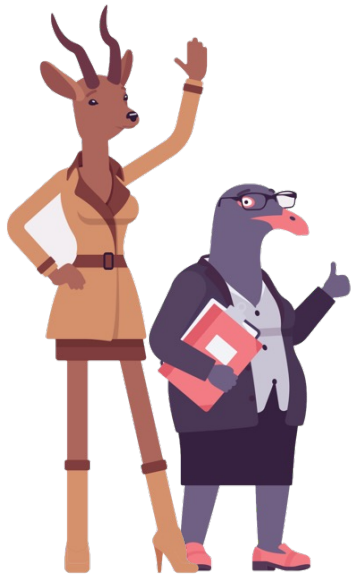
The cloud focused here first.

Doesn't take much power

Lots of these can coexist on the same hardware

Backup, restore, CHECKDB are easy and fast





That's why Alex & Billie love the cloud.

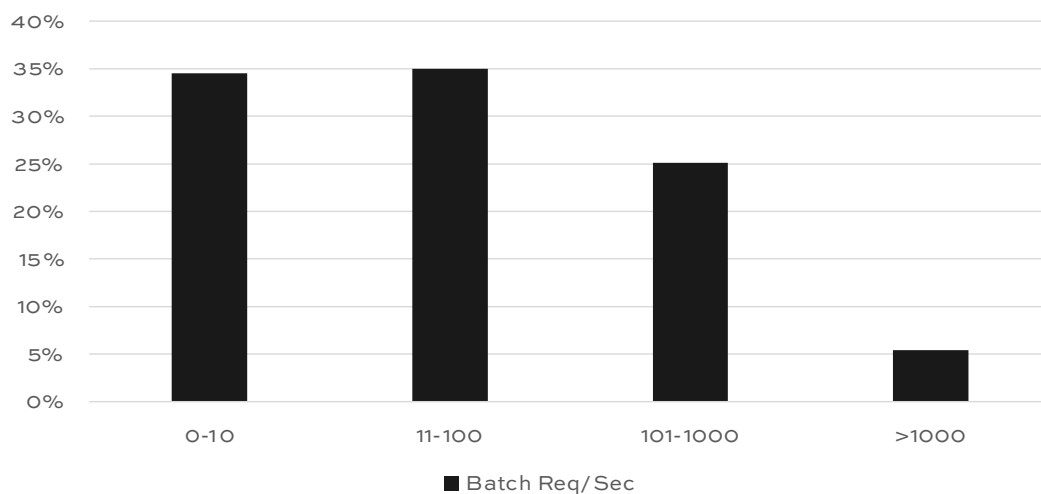
Their databases tend to be <1 TB.

Their databases tend to be standalone.

Deck 1 p29



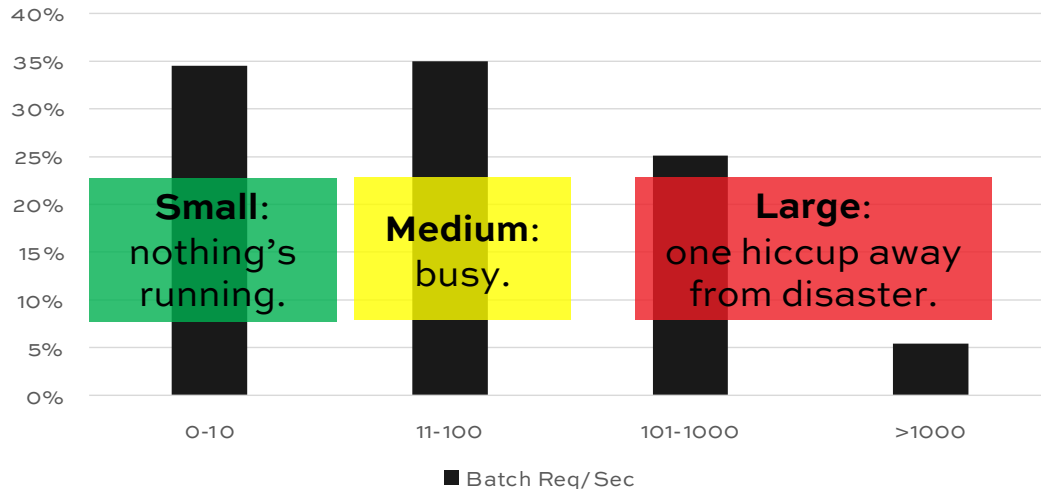
2. Batch Requests per Second



Deck 1 p30



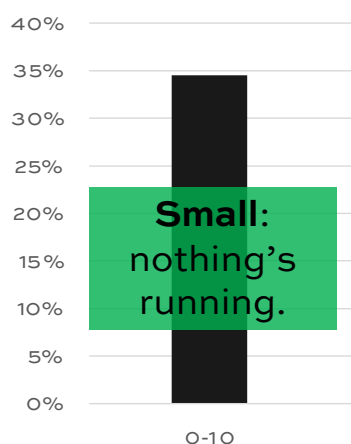
2. Batch Requests per Second



Deck 1 p31



2. Batch Requests per Second



The cloud focused here first.

Easy to find maintenance windows

Few transaction log issues

Easy to move from one hardware box to another

Doesn't need a full time DBA

Deck 1 p32



Billie the Small Biz Bird fits great.

Small businesses don't have that many simultaneous users.

Their web sites tend to be hosted elsewhere, like SquareSpace, not on SQL Server.

They tend to not have active e-commerce sites.



Deck 1 p33



Alex the Architect may not.

Alex's apps may be web-facing.

Even if Alex put each client in its own database, Alex may worry that a single big client might run over 10 queries per second.

That's fine – Alex can still migrate to the cloud.

But Alex's database may not be all green here.



Deck 1 p34



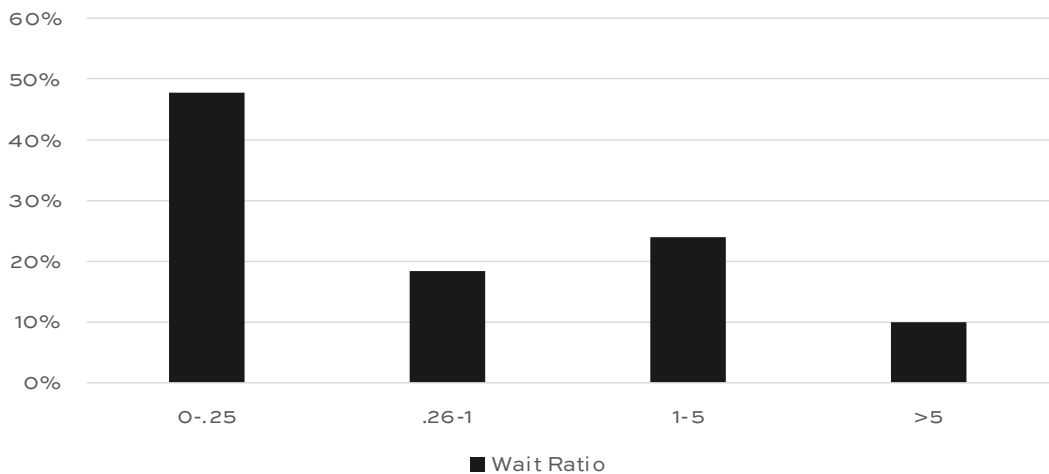
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Deck 1 p35



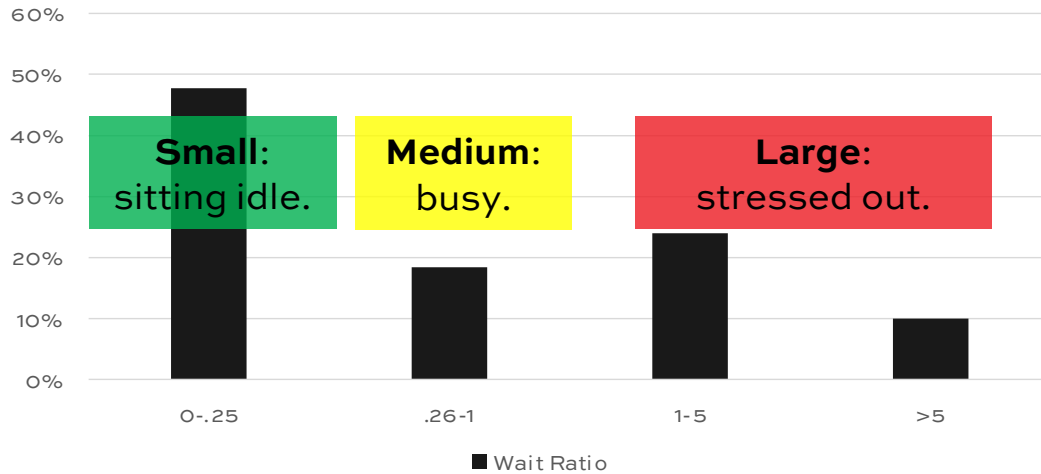
3. Wait Time Ratio



Deck 1 p36



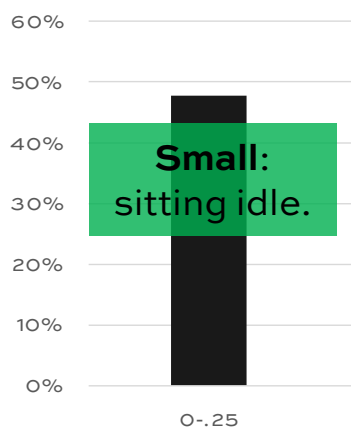
3. Wait Time Ratio



Deck 1 p37



3. Wait Time Ratio



The cloud focused here first.

Easy to find maintenance windows

Low hardware requirements

Profitable for MS to host

Doesn't need a full time DBA

Deck 1 p38



Anybody could be green or red.



Summary

	Small	Medium	Large
1. Total size, GB	0-250	250-1TB	>1TB
2. Batch req/sec	0-10	10-100	>100
3. Wait time ratio	0-0.25	0.25-1	>1
PaaS will work	Yes	With some work	Maybe not
IaaS VMs will work	Yes	With some work	Maybe not

Deck 1 p40

These aren't guarantees.

Some small, low-load databases won't work if they rely on features you can't get in the cloud.

Some huge, high-load databases work great if they're well-tuned and closely monitored.

Today is about understanding the gotchas.

Deck 1 p41



3 Questions for Stakeholders



Deck 1 p42



1. Are you content with the app's current performance?

It's often no, so here's another way to phrase it:

“If we migrated this application to the cloud and we got the same performance we get today, would this project be a success or a failure?”

Deck 1 p43



What their answers mean

If they say the same performance is a **success**, the hardware can be the same size, and maybe even smaller to cut costs.

If they say the same performance is a **failure**:

- We may have to add hardware
- We may have to tune indexes
- We may even have to tune code

Deck 1 p44



Know your target audience.

Some companies go to the cloud to cut costs, and they're willing to cut performance on legacy apps.

Some companies try to gain performance because they were stuck on old legacy hardware, and they're ready to move to newer/faster/bigger.

Deck 1 p45



2. How much growth do we expect this year?

Some companies migrate to prep for rapid scale, and expect the new servers to be flexible, quickly.

Some migrate to cut hosting costs for legacy apps, and the new replacement apps will be cloud-native.

Some just migrate to escape old data centers, and expect no capacity changes.

Deck 1 p46



What their answers mean

Your solution may need to include things like:

- An Always On Availability Group with the ability to quickly add bigger or smaller replicas, and fail over to them to handle load change, or offload reads
- A single VM that will be gradually scaled down over time as the load dies off

Deck 1 p47



3. Can we contain our developers to only Azure SQL DB features?

Single database (no cross-db queries, transactions)

No FILETABLE, FILESTREAM

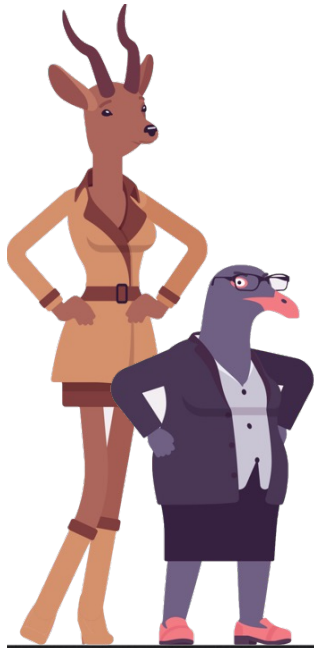
No OPENQUERY

No fancy replication

No SQL Agent jobs

Deck 1 p48





What their answers mean

If your devs can stick with Azure SQL DB features, then you can use Azure SQL DB.

If you can... you probably should.

It helps Alex the Architect and Billie's Small Business postpone hiring full time DBAs.

Deck 1 p49



Now, time to reflect.

Answers you gathered

1. Total data size in GB/TB, number of databases
2. Batch Requests per second
3. Wait time per core over time (like per second)
4. Is the same performance a success or a failure?
5. How much growth do we expect this year?
6. Can we stick to Azure SQL DB features?

These help us be more confident about:

- The kind of workload testing we need
- The cloud infrastructure size range choices

Deck 1 p51



Small? Users happy? No expected growth?

	Small
1. Total size, GB	0-250
2. Batch req/sec	0-10
3. Wait time ratio	0-0.25
PaaS will work	Yes
IaaS VMs will work	Yes

You may not need any load testing.

You might even be able to downsize from your current hardware.

Deck 1 p52



Large? Users unhappy? Growth coming?

1. Total size, GB
2. Batch req/sec
3. Wait time ratio
PaaS will work
IaaS VMs will work

Large
>1TB
>100
>1
Maybe not
Maybe not

Deck 1 p53



Beware.

Recap: 3 metrics + 2 questions

1. Total data size in GB/TB, number of databases
2. Batch Requests per second
3. Wait time per core over time (like per second)
4. Is the same performance a success or a failure?
5. How much growth do we expect this year?
6. Can we restrain ourselves to Azure SQL DB features?

These predict the ease of cloud migrations.

